

Determinants of Current Account Balances of Non-Oil Developing Countries in the 1970s

An Empirical Analysis

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THE PAST DECADE PROVED to be a period of considerable stress for non-oil developing countries. Throughout most of the 1970s, a combination of events caused the international economic environment to become less conducive to stable growth for this group of countries and made the problem of economic management in general—and of balance of payments adjustment in particular—much more difficult. The substantial fluctuations in the world market prices of primary commodities, the sharp increases in the price of energy products, the slowdown of economic activity in the industrial countries, and the rise in real interest rates toward the end of the period were all major contributors to a serious deterioration in the current account positions of most non-oil developing countries. At the same time, domestic developments in a number of economies also played a significant role in exacerbating payments disequilibrium. In many non-oil developing countries, inflationary demand-management policies—combined with rigid exchange rate policies and restrictions on trade and payments—resulted in domestic demand pressures and cumulative losses in international competitiveness that also gave rise to current account and overall balance of payments difficulties.

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While the broad outlines of these developments have been discussed at length in the literature, the assessment of the contributions of the afore-mentioned factors to the payments problems of developing countries has often relied on casual observation, rather than on a systematic evaluation of trends in a broad-based sample of non-oil developing countries. A number of studies, including Reichmann (1978), Dell (1980), Dell and Lawrence (1980), Killick (1981), and Khan and Knight (1982), have drawn conclusions from the "stylized facts" of developing countries' experience during the past decade but have not subjected the available data to standard empirical tests. The purpose of this paper is to go beyond these previous studies and to examine empirically the influences of external and domestic factors on the evolution of the current accounts of non-oil developing countries during the 1970s. For this purpose, a simple model is specified that relates the current account to its main determinants and the relationship is estimated for a broad group of 32 non-oil developing countries. The results of this exercise are then used to draw inferences about the relative contributions of various factors to the behavior of the current accounts of the countries in this group during the period 1973–81—a matter over which there is still considerable controversy.¹ It is argued here that this continuing controversy on the role of external and internal factors stems to a large extent from the lack of formal statistical testing of the basic relationships involved.

At the outset, it is necessary to point out certain areas that the paper does not cover, even though they are closely related to the subject at hand. First, there is no explicit consideration of the important question of how the burden of external adjustment should be shared among surplus and deficit countries, or between the non-oil developing countries as a group and the industrial world. These are essentially normative issues about how the international monetary system should ensure some degree of symmetry between various countries in undertaking balance of payments adjustment. As such, they extend beyond the scope of the empirical analysis undertaken here. Second, the issue of the appropriate trade-off between adjustment and financing in the con-

¹For example, Dell (1980) argues that most of the deterioration in the current account balances of developing countries during the period 1973–76 can be attributed to external factors, and principally to adverse changes in the terms of trade. This has been disputed by, among others, Killick (1981) and Khan and Knight (1982).

text of transitory versus permanent shocks to the balance of payments is covered only in passing. A number of recent papers (Nowzad (1981), Guitián (1981), and Polak (1982)) have dealt extensively with this particular topic.

The outline of the rest of the paper is as follows. Section I briefly describes recent current account developments in the non-oil developing countries and discusses the behavior of the various factors considered responsible for these developments. Section II assesses the quantitative role of the main factors on the basis of empirical tests undertaken with a pooled cross-section time-series sample of the 32 non-oil developing countries for which the necessary data are available. Section III briefly summarizes the results and indicates their relevance for balance of payments adjustment policies in developing countries.

I. Factors Affecting Current Account Positions

The combined current account deficit of non-oil developing countries, expressed as a proportion of their exports of goods and services, rose sharply from an average of about 17 percent during the period 1967–73 to more than 20 percent in the period 1974–81 (Table 1). During the latter period, there were also considerable year-to-year fluctuations in this ratio. In the aftermath of the first oil price increase in 1973–74, there was a sizable worsening of the current account positions of non-oil developing countries, with their combined deficit reaching a peak of nearly 31 percent of exports of goods and services in 1975. This percentage represented a near tripling from the value registered in 1973. Favorable movements in the prices of primary commodities led to a marked improvement in the current account balance in 1976–77, but from 1978 onward the ratio of the combined deficit to exports of goods and services continued to rise steadily at the rate of approximately 2 percentage points per annum. The second round of oil price increases in 1979–80 appears to have had a much smaller impact on the current accounts of non-oil developing countries than did the earlier increase; indeed, there was no important difference between the rate at which the combined current account deficit increased in the year preceding the oil price increase (1978) and the rate during the two years that followed it (1980–81).

The factors that have typically been identified as having exerted an important influence on the current account positions of the

TABLE 1. NON-OIL DEVELOPING COUNTRIES: CURRENT ACCOUNT BALANCES, CHANGES IN TERMS OF TRADE, AND FOREIGN REAL INTEREST RATES, 1973-81
(In percent)

	1973	1974	1975	1976	1977	1978	1979	1980	1981
Current account balance ¹	-10.6	-24.8	-30.9	-18.1	-12.9	-15.2	-17.7	-20.4	-22.4
Changes in terms of trade ²	6.7	-5.1	-10.0	5.6	6.9	-5.4	0.8	-6.4	-8.1
Foreign real interest rate ³	-17.9	-20.1	8.1	-1.4	-7.1	3.3	-4.7	-4.6	17.4
Memorandum item									
Real GNP growth of industrial countries	6.2	0.6	-0.5	4.9	4.0	4.0	3.6	1.3	1.2

Sources: International Monetary Fund, *World Economic Outlook*, IMF Occasional Paper, No. 9 (Washington, April 1982), and *International Financial Statistics* (various issues).

¹As a percentage of total exports of goods and services.

²Annual percentage change.

³Eurodollar deposit rate adjusted for changes in an index of export prices of non-oil developing countries (expressed in U.S. dollars).

non-oil developing countries during the 1970s are as follows: (1) the deterioration in the terms of trade; (2) the slowdown of economic activity in the industrial countries; (3) the sharp increase in the level of real interest rates in international credit markets, particularly toward the end of the decade; and inadequate or insufficient domestic adjustment evidenced by (4) rising fiscal deficits and (5) appreciation of real effective exchange rates. This list is certainly not exhaustive, but it does cover the more important causes of the current account difficulties experienced by most non-oil developing countries.² These five factors have sometimes been divided into "external" and "domestic" categories (Dell (1980) and Killick (1981)), although this distinction is perhaps not particularly useful in practice, owing to the close interrelationships that often exist among different factors. Nevertheless, for expositional purposes, it may be convenient to view the first three factors—namely, variations in the terms of trade, the growth rates of industrial countries, and foreign real interest rates—as "external," in the sense that these are effectively exogenous to the typical non-oil developing country. By analogous reasoning, changes in fiscal deficits and in real effective exchange rates can be treated as "domestic" or endogenous factors, to the extent that national authorities control public sector revenue and expenditures, and their domestic economic policies influence both the nominal exchange rate and domestic input and product prices. We now turn to a brief discussion of each of these external and domestic factors.

EXTERNAL FACTORS

Terms of trade

The terms of trade of non-oil developing countries taken as a group fell at an average rate of about 2 percent a year over the period 1973–81 (Table 1).³ During the preceding ten-year period (1963–72), the terms of trade of this group had improved at an average rate of $\frac{1}{2}$ of 1 percent a year, so that the decline that began in 1973 represented a distinct change from the historical

²The analysis abstracts, therefore, from the effects of domestic supply shocks, for example, droughts and other weather-related phenomena and rising protectionism in the export markets of developing countries that were also important elements in the experiences of a number of non-oil developing countries.

³The terms of trade are defined in the customary manner as the ratio of the price of exports to the price of imports, expressed in U.S. dollars.

trend. A considerable part of the terms of trade deterioration in the 1970s can be attributed to the rise in import prices that resulted from the fourfold jump in the world price of energy products in 1973–74 and the further substantial increase that occurred in 1979–80. For example, Cline (1981) found that, whereas 15 of the 25 non-oil developing countries in his sample experienced a deterioration in their terms of trade between 1971–72 and 1975, 19 of these countries showed an improvement once the cost of oil imports was excluded from their import price data.

The declines in the terms of trade that occurred after each episode of oil price increases were broadly similar—7.6 percent (1974–75) and 7.3 percent (1980–81). In both cases, favorable movements in the prices of primary commodities coincided with the oil price increase and helped to mitigate part of the adverse effect. For example, the price index of non-oil primary commodities rose by 28 percent in 1974; during 1979–80 it registered an average increase of about 12 percent a year. While there were years when commodity prices fell, such as 1975 (by 18.2 percent), 1978 (by 4.7 percent), and 1981 (by 14.8 percent), the average annual rate of increase was nearly 12 percent for the period 1973–81 as a whole. Furthermore, it is possible that the increase would have been even larger had growth rates in industrial countries not decelerated perceptibly toward the end of the 1970s. Evidence on the association between economic activity in industrial countries and the prices of primary products was provided by Goreux (1980) in a study of the International Monetary Fund's compensatory financing facility. For a sample of 37 primary commodities, Goreux found that over the period 1962–79, fluctuations in primary commodity prices could be explained to a large extent by cyclical movements in economic activity in industrial countries and by world inflation. The empirical results indicated that each change of 1 percent in the business cycle index for industrial countries tends to be associated with a change of 2.2 percent in the same direction in the prices of primary commodities.

A point worth mentioning in the context of these terms of trade developments is that the predicament of non-oil developing countries was not unique. The terms of trade of industrial countries on average fell by more (2.3 percent a year) during 1973–81 than they did for non-oil developing countries (1.7 percent a year). This fact is perhaps not too surprising, since petroleum products represent a relatively larger share in the total imports of industrial countries than they do in those of non-oil developing countries. Of course,

it is true that, for a variety of reasons, industrial countries were in a better position to adjust to the deterioration of their terms of trade than were non-oil developing countries, so that the impact on the latter group was generally more severe. The only gainers in the 1970s were, as one would expect, the group of oil exporting countries, which experienced an average improvement of about 25 percent per annum in their terms of trade.

Broadly speaking, there appears to be sufficient synchronous movement in the current account ratios and terms of trade changes during the period under consideration to allow one to conclude that there is an association between the two variables (Table 1). While a more detailed examination of this relationship is undertaken later in the paper, it may be useful to present a preliminary notion of the statistical nature of the relationship on the basis of the aggregative data shown in Table 1. A simple regression of the annual change in the combined current account position of the non-oil developing countries (expressed as a proportion of their exports of goods and services (ΔCA)) on the percentage change in their terms of trade ($DTOT$) and a linear time trend (T) yields the following results:⁴

$$\Delta CA_t = -6.352 + 0.851 DTOT_t + 1.237T \quad (1)$$

(1.51) (2.66) (1.57)

$$R^2 = 0.408; D-W = 1.65$$

While these estimates are based on a small number of observations (9), it is nevertheless worth noting that the coefficient measuring the effect of movements in the terms of trade on the change in the current account ratio has the expected positive sign and is significantly different from zero at the 5 percent level. These results are suggestive, although they cannot by any means be treated as definitive.

Slowdown of growth in industrial countries

Apart from the indirect effect working via changes in the terms of trade of non-oil developing countries, growth in industrial countries also has a more direct impact on current accounts through its influence on the exports of non-oil developing countries. There was a pronounced decline in the average growth rate of the real gross national product (GNP) of industrial countries

⁴T-values are shown in parentheses below the coefficients. R^2 is the adjusted coefficient of determination, and $D-W$ is the Durbin-Watson test statistic.

between the period 1963–72 (4.7 percent) and the period 1973–81 (2.8 percent); for the subperiod 1979–81, the average annual growth rate was only 2 percent. Growth in the volume of exports of non-oil developing countries also fell, but the decline was a relatively modest one—from 6.7 percent in the period 1963–72 to 5.9 percent in the years 1973–81. Whereas the average growth rate of imports of industrial countries fell quite sharply (from 9 percent (1963–72) to 3.6 percent (1973–81)), this was apparently not reflected in a proportionate decline in export growth for non-oil developing countries as a group.

In a recent paper, Goldstein and Khan (1982 a) have argued that during the period 1973–81 two main factors helped to minimize the consequences of this slower growth of industrial countries' imports for the exports of non-oil developing countries. First, non-oil developing countries, particularly those with a relatively higher proportion of manufactures in their total exports, were able to capture a larger share of the industrial countries' slow-growing import volume. The process was assisted in the beginning of the period by the granting of tariff preferences but was partially reversed later, as protectionist pressures intensified in the industrial world. Second, non-oil developing countries were able to increase their total exports (in volume terms) faster than their exports to industrial countries by directing a larger share to oil exporting countries, which were becoming increasingly important markets for exports of both manufactures and primary products.

Foreign real interest rates

The third major external factor affecting the current accounts of non-oil developing countries, particularly during the late 1970s, was the sharp increase in service payments on external debt. Debt service had not been a very serious problem for many non-oil developing countries during most of the period prior to 1975 because conditions in the international credit markets were generally favorable, and a large proportion of outstanding debt, particularly for the low-income countries, had been lent by foreign official institutions during the 1960s at fixed concessionary rates. As a result, the effective nominal interest rates on the external debt of many non-oil developing countries, when adjusted by the increase in their export prices, yielded real interest rates that were low or negative during this period (Artus (1983)). This conclusion is substantiated by the data on foreign real interest rates presented

in Table 1, which shows that for most of the period 1973–77 real interest rates on foreign debt were decidedly negative, averaging about -7.7 percent.

In 1978 this picture changed quite drastically. Owing to adverse terms of trade shocks and weakness in export market growth, the non-oil developing countries' stock of external debt, particularly short-term debt, rose sharply. In addition, interest rates in international capital markets were climbing to postwar highs at a time when developing countries' export prices began to weaken. Real interest rates on external debt became positive and averaged about 3 percent a year during the period 1978–81—a turnaround of some 11.5 percentage points (Table 1). The high real interest rates that generally prevailed after 1978 exerted their strongest impact on the debt service burdens of those countries whose stocks of external debt were relatively large; these were obviously the countries that had already experienced substantial current account deficits and had resorted to foreign financing in earlier years. From this point of view, the recent rise in debt service burdens can also be seen at least partly as the lagged effect on the current account of the other factors that are discussed in the present paper.

DOMESTIC FACTORS

Domestic demand pressures have historically been an important factor affecting the current account positions of non-oil developing countries. Evidence on this issue has been provided by Reichmann (1978), Dell and Lawrence (1980), and Killick (1981). For example, in examining the causes of balance of payments problems for 21 countries that had stand-by arrangements with the Fund during the period 1973–75, Reichmann (1978) concluded that overexpansionary demand policies were the major factor in 15. On the other hand, Dell (1980, p. 834) has argued that over a similar period (1973–76) "... demand pressures emanating from domestic economies were far less important, relative to other causes of change in the trade balance, than had previously been the case."

Excess domestic demand generally manifests itself in a worsening of the balance of payments and a rise in the domestic inflation rate, with the relative sizes of these effects depending in large part on the openness of the economy. While the causes of excess demand can be many, it is argued here that in non-oil developing countries the rise in aggregate demand can often be traced back

to expansionary government policies that result in fiscal deficits. Another domestic factor that is closely associated with such fiscal deficits, and which operates through the rise in domestic inflation, is an appreciation of the real effective exchange rate. During the 1970s, both rising fiscal deficits and real effective exchange rate appreciations frequently combined to have serious consequences for the current account balances of non-oil developing countries.

Fiscal deficits

There is now considerable evidence available on the existence of a positive relationship between fiscal deficits and current account deficits. The most direct way in which the two are related can be observed within the framework of the national income identity that links the current account to the gap between total savings and investment. Other things being equal, an increase in the fiscal deficit (dissaving by the public sector) tends to raise domestic absorption and thereby worsen the current account.⁵ Furthermore, in non-oil developing countries, the fiscal deficit is generally matched by a corresponding rise in domestic liquidity, which then expands private nominal demand and reinforces the negative impact on the current account.

The fiscal position of non-oil developing countries, measured as a proportion of gross domestic product (GDP),⁶ was consistently in deficit throughout the period 1973–81. The deficit rose from about 2 percent in 1973 to a little more than 3 percent in the period 1975–76. After a slight improvement in 1978 there was a significant worsening, and the ratio of the fiscal deficit averaged about 3½ percent during the years 1979–81. The steady rise in the fiscal deficit was evidently associated with the worsening current account positions of non-oil developing countries, and it is hypothesized that it was a major domestic cause of current account problems.

At the same time, inflation was also endemic in most non-oil developing countries during the period 1973–81, averaging about 29 percent a year for the group, compared with about 12 percent in the period 1963–72. This general increase in the overall inflation rate can also be attributed in large part to the direct pressure

⁵This type of analysis is the mainstay of the so-called fiscal approach to the balance of payments. See Kelly (1982) for a brief summary of this approach.

⁶The data utilized here are based on a sample of 100 non-oil developing countries. See IMF (1982).

on available resources exerted by an increase in government demand, as well as the effect working through the expansion in the money supply that resulted from the financing of increased fiscal deficits. In the typical non-oil developing country, the absence of a well-developed domestic capital market precludes the sale of substantial amounts of public debt to the nonbank sector, so that the government has to rely on borrowing from the banking system or increasing its foreign debt to meet its financing requirements. As a result of such methods of financing, the fiscal deficit will lead directly to increases in the monetary base and will tend to create inflationary pressures. The 1970s were certainly characterized by rapid monetary growth associated with the financing of fiscal deficits, and this undoubtedly played a significant role in the inflationary process.

Real effective exchange rates

More important, however, from the point of view of current account developments, there was a tendency for nominal exchange rate changes in many developing countries to be less than would have been needed to offset the excess of domestic inflation rates over inflation in the rest of the world,⁷ and in general to lag behind changes in the domestic price level, resulting in an appreciation in the real effective exchange rate.⁸ This tendency would imply the existence of a positive relationship between domestic demand pressures, inflation, and the real effective exchange rate; such a link has been documented by Aghevli (1982) for a number of Asian countries and by Khan and Knight (1982) and IMF (1982) for broader groups of developing countries. The behavior of the real exchange rate, being the outcome of changes in the nominal exchange rate and domestic inflation, essentially reflects the way in which exchange rate policy and demand-management policies are mutually coordinated.

An increase in the real effective exchange rate is clearly a fundamental determinant of the deterioration in a country's current account, since, other things being equal, it tends to raise the demand for imports and to reduce foreign demand for exports. Furthermore, if the price of exports is fixed exogenously in world

⁷Allowing for any equilibrium changes in national price levels. See Frenkel and Mussa (1981).

⁸For purposes of this exercise, the real exchange rate is defined as the home country's consumer price index relative to an import-weighted average of consumer price indices in partner countries, adjusted for the nominal exchange rate.

markets while domestic nominal wages rise in line with domestic prices, an appreciation in the real exchange rate induces a cost squeeze on the exporting sector that reduces the supply of exportables. On the basis of these effects, it seems legitimate to view movements in the real effective exchange rate, along with the fiscal position, as useful summary indicators of the domestic factors that typically would be expected to influence the current account. At the same time, it should also be kept in mind that external factors, such as changes in the terms of trade, may also exert a systematic influence on the real effective exchange rate, so that it is not always a reflection of domestic factors alone.⁹ It is this type of interaction that gives rise to practical difficulties in making a clear-cut distinction between so-called external and internal factors.

II. Empirical Estimates of Factors Affecting Current Account Balances

The analysis so far has been both descriptive and aggregative in nature, and in this section a more systematic empirical examination is made of the respective influences of the five factors discussed in Section I on the evolution of current account positions. For this purpose, the evidence is examined for 32 non-oil developing countries for which the relevant published data are available for the period 1973–80.¹⁰ The purpose here is to test the influences of each of the five factors on the current accounts of this group of non-oil developing countries. To do so, a simple model of the current account is formulated and estimated that introduces the external and domestic factors independently as the principal explanatory variables.

The basic current account equation considered here has the following general form:

⁹For example, a worsening of the terms of trade owing to an increase in import prices would raise the domestic price level. If domestic policies, including exchange rate policy, were not changed, the real effective exchange rate, as defined here, would tend to appreciate.

¹⁰These 32 countries are Bolivia, Brazil, Burma, Colombia, Cyprus, Dominican Republic, Ecuador, Ethiopia, Fiji, Greece, Guyana, Honduras, Israel, Jamaica, Jordan, Kenya, Korea, Malawi, Malaysia, Malta, Mauritius, Pakistan, Panama, Paraguay, Philippines, Rwanda, South Africa, Sri Lanka, Suriname, Thailand, Turkey, and Yugoslavia. Since the requisite published data are not available for all these countries for 1981, the period of coverage was reduced to 1973–80.

$$CA/X = f(TOT, DYIC, RRI, RER, FP/Y, T) \quad (2)$$

where

- CA = current account balance (excluding official transfers)
 X = nominal exports of goods
 TOT = terms of trade
 $DYIC$ = growth of real GNP in industrial countries
 RRI = foreign real interest rate
 RER = real effective exchange rate index
 FP = fiscal position (revenues *minus* expenditures)
 Y = nominal GDP
 T = linear time trend

Based on the discussion in Section I, one would expect an improvement in the terms of trade or an increase in the growth rates of industrial countries to result in an improvement in the current account, while a rise in the foreign real interest rate or an appreciation in the real effective exchange rate would tend to worsen it.¹¹ The time trend variable is assumed to capture the effects of other factors, both external and domestic, on the current account.¹²

In a sense, equation (2) can be viewed as an unrestricted reduced form relationship that is derived in a straightforward manner from a structural model of the components of the current account—imports, exports, and net service payments. While it could certainly be argued that this formulation is excessively simple and may not include all the relevant determinants of the current account balance, the estimates of equation (2) should still yield useful information on the specific question that is of concern here, namely, the relative importance of the various factors. For estimation purposes, the following specific form of (2) was utilized:

$$(CA/X)_t = a_1 \log TOT_t + a_2 DYIC_t + a_3 RRI_t + a_4 \log RER_t + a_5 (FP/Y)_t + a_6 T \quad (3)$$

¹¹The export variable (X) is used only to scale the current account balance to make it comparable across countries. To avoid problems associated with exchange rate conversions, it was considered preferable not to use domestic income to scale the dependent variable.

¹²Presumably, this would also include the effects of rising protectionism over the sample period.

Certain restricted versions of equation (3) were estimated—namely, one relating the current account ratio only to external factors, equation (3.1), and then only to the two domestic factors, equation (3.2). The most general equation (3.3) includes both sets of factors as explanatory variables.¹³ Estimating these three equations should prove to be sufficient to isolate the factors that have exerted the strongest influence on the dependent variable during the sample period.

The three versions of equation (3) were estimated using pooled time-series cross-section data for the sample of 32 non-oil developing countries. The data for each country comprised eight annual observations over the period 1973–80.¹⁴ Allowance was made in the specification for cross-country differences in the ratio of the current account to exports during the sample period by the addition of 32 country dummies. It was also explicitly assumed that the parameters ($a_1, \dots, a_6$) were the same across countries, so that no slope dummies were introduced. As a consequence, the estimates of the equations should be interpreted as being relevant for an “average” or “typical” non-oil developing country, rather than applying to any specific country.

The results for the three versions of the basic equation are presented in Table 2. In the equation containing only external factors (equation (3.1)), all three economic variables yield coefficients that have the expected signs and are significantly different from zero at the 5 percent level. A deterioration in the terms of trade does indeed result in a worsening of the ratio of the current account to exports, as does the decline in the growth rate in industrial countries in the 1970s. The effect of the latter variable is perhaps not as strong as one might have expected; this probably reflects the fact that non-oil developing countries were partially successful in offsetting the direct effects of the slowdown in industrial countries by curtailing imports and, to a lesser degree, by increasing their exports to other regions, particularly the oil ex-

¹³In all cases, the trend variable was included.

¹⁴This yielded 256 observations for each of the variables. Basic data are obtained from International Monetary Fund, *International Financial Statistics* (various issues). For each country, the variables CA and X are in current U.S. dollars; TOT is the ratio of the unit value of exports to the unit value of imports, both expressed in terms of U.S. dollars; RRI is the three-month Eurodollar deposit rate adjusted for changes in the individual country's U.S. dollar export price index; RER is calculated using 1977 import weights and the relevant consumer price indices; and FP/Y is the ratio of government revenues *minus* expenditures to nominal GDP.

TABLE 2. THIRTY-TWO NON-OIL DEVELOPING COUNTRIES: RESULTS FOR CURRENT ACCOUNT BALANCES, 1973-80¹

Equation Number	Terms of Trade (log <i>TOT</i>)	Growth in Industrial Countries (<i>DYIC</i>)	Real Foreign Interest Rate (<i>RRl</i>)	Real Effective Exchange Rate (log <i>RER</i>)	Fiscal Position (<i>FP/Y</i>)	Time Trend (<i>T</i>)	<i>R</i> ²	<i>SEE</i>
(3.1)	0.436 (3.61)	1.767 (2.22)	-0.459 (3.75)			-0.011 (1.43)	0.901	0.275
(3.2)				-0.561 (2.91)	1.645 (2.83)	-0.029 (3.57)	0.907	0.267
(3.3)	0.532 (4.47)	1.452 (1.86)	-0.353 (2.93)	-0.597 (3.34)	1.483 (2.80)	-0.014 (1.88)	0.913	0.260

¹Dependent variable is the ratio of the current account to merchandise exports. *T*-values are presented in parentheses below the coefficients, *R*² is the adjusted coefficient of determination, and *SEE* is the standard error of the estimated equation.

porting countries. In addition, since the indirect effect of a change in industrial-country growth performance on the non-oil developing countries is already captured to some extent in the estimate of a_1 , the estimate of a_2 probably does not represent the total impact exerted by variations in industrial-country growth on the current account balances of non-oil developing countries. Despite the fact that variations in the foreign real interest rate are generally considered to have become quantitatively important only toward the end of the period, the coefficient of this variable turns out to be rather precisely estimated and has the sign that one would expect—the increase in the foreign real interest rate appears to have added significantly to the worsening of the current account of non-oil developing countries.¹⁵

Both of the domestic variables—the real effective exchange rate and the ratio of the fiscal position to GDP—also appear to be important in explaining current account developments in the period 1973–80 (equation (3.2)). As hypothesized, an increase in the real effective exchange rate or a deterioration in the fiscal position¹⁶ has a negative impact on the current account ratio, and the coefficients are significantly different from zero at the 5 percent level. The fit of equation (3.2) is fairly close to that obtained for equation (3.1), so that it is difficult to choose between the two specifications on purely statistical criteria.

The most general equation (3.3) provides empirical confirmation for the view that both external and internal factors were at play in affecting the current account outcome for non-oil developing countries. All the coefficients have the expected signs, and with the exception of the coefficient for the growth rate of industrial countries, are all significant at the 5 percent level.¹⁷ Clearly, the five factors that have been considered are not the only ones that are important, since the time trend variable, which is intro-

¹⁵Ideally, one would wish to scale the foreign real interest rate faced by each non-oil developing country by its outstanding stock of foreign debt, to reflect changes in interest payments more accurately. This was not possible, owing to the absence of data on *total* stocks of foreign debt for the individual countries in the sample. In effect, the simple specification assumes implicitly that the stock of foreign debt has grown smoothly over the period, so that most of the year-to-year variations in interest payments have occurred because of changes in the interest rate on this debt.

¹⁶As the variable FP/Y is defined as the ratio of the difference between government revenues and expenditures to GDP, an increase in FP/Y implies an improvement in the fiscal position, and vice versa.

¹⁷In this equation, however, the coefficient for industrial-country growth is significantly different from zero at the 10 percent level.

duced to take account of those influences that are not explicitly included in the specification, yields a coefficient that is significant at the 10 percent level. The negative sign of this trend coefficient indicates that the factors not included explicitly in the specification exerted an additional systematic negative influence on the current account balances of non-oil developing countries during the period 1973–80.¹⁸ By and large, equation (3.3) appears to be reasonably well determined and explains a large proportion of the variance in the current account ratios in this sample.

Given the units in which the variables are expressed, the estimates suggest that an increase of 1 percentage point in the ratio of the average non-oil developing country's fiscal deficit to its GDP would cause its current account, as a proportion of its exports, to deteriorate by about 1½ percentage points. A deterioration of 1 percent in the external terms of trade or an appreciation of 1 percent in the real effective exchange rate would lead, on average, to a decline of about ½ of 1 percentage point in the current account ratio. Finally, a fall of 1 percentage point in the growth rate of industrial countries would reduce the current account ratio by a little less than 1½ percentage points, while a similar rise in the real foreign interest rate would induce a fall of less than ½ of 1 percentage point.

As the units of measurement differ for the variables, an alternative way of ascertaining the relative influence of the factors under consideration is to calculate the relevant Beta coefficients. As is well known, Beta coefficients measure the change in the explained variable (in standard-deviation units) for a unit change in each explanatory variable (also expressed in standard-deviation units), holding all other variables constant. The Beta coefficients are independent of the units of measurement of each explanatory variable and can thus be compared directly. These coefficients, along with their respective standard errors, are presented in Table 3.¹⁹

It is evident from Table 3 that the most important explanatory variable is the terms of trade, while the least important factors are growth in the industrial countries and the time trend. However, as was mentioned earlier, it is possible that some part of the effect

¹⁸As was mentioned earlier, rising protectionism in the export markets of these countries could be one possible candidate, among others.

¹⁹Since the statistical distribution of Beta coefficients is unknown, one cannot perform formal tests of significance in assessing the relative importance of the external and domestic factors.

TABLE 3. THIRTY-TWO NON-OIL DEVELOPING COUNTRIES:
VALUES AND STANDARD ERRORS OF BETA COEFFICIENTS
IN EQUATION (3.3), 1973–80

Variable	Beta Coefficient	Standard Error
$\log TOT$	0.179	0.04
$DYIC$	0.046	0.03
RRI	-0.084	0.03
$\log RER$	-0.112	0.03
FP/Y	0.119	0.04
T	-0.048	0.03

of the slowdown in the growth rate of the industrial countries is being picked up by variations in the terms of trade, and also perhaps the foreign real rate of interest. The foreign real interest rate, the real effective exchange rate, and the government's fiscal position turn out to be of roughly equal importance as determinants of current account developments in this particular sample of non-oil developing countries.

While the fit of the model for the entire pooled sample is quite good, it would be useful to ascertain how well equation (3.3) does in explaining the current account ratio for each individual country. One simple way of doing this is to obtain country-specific measures of goodness-of-fit by calculating the correlation between the actual and fitted values of the current account to exports ratio using the observations pertaining to each country in the pooled sample. The purpose of this calculation is to judge the extent to which the overall conclusions are influenced by the stringent simplifying assumption that is used to obtain the pooled-sample estimate; namely, that a given change in one of the explanatory variables exerts the *same* quantitative impact on the current account ratio for each of the 32 countries. The relevant correlation coefficients, calculated for each country from the estimates of equation (3.3), are presented in Table 4.

The results of this exercise are quite interesting in that they suggest only a few countries for which the simplifying assumption referred to earlier is likely to result in serious in-sample prediction errors. For three countries—Malta, Panama, and Suriname—the correlation coefficients are obviously poor, but for the other 29 countries the coefficient is generally above 0.5 and in 21, it is more

TABLE 4. THIRTY-TWO NON-OIL DEVELOPING COUNTRIES: COEFFICIENTS OF CORRELATION BETWEEN ACTUAL AND PREDICTED VALUES OF CURRENT ACCOUNT BALANCES, 1973-80

Country	Correlation Coefficient
Bolivia	0.693
Brazil	0.772
Burma	0.712
Colombia	0.715
Cyprus	0.637
Dominican Republic	0.850
Ecuador	0.886
Ethiopia	0.843
Fiji	0.667
Greece	0.470
Guyana	0.805
Honduras	0.759
Israel	0.925
Jamaica	0.559
Jordan	0.623
Kenya	0.776
Korea	0.731
Malawi	0.750
Malaysia	0.735
Malta	0.009
Mauritius	0.976
Pakistan	0.869
Panama	0.299
Paraguay	0.985
Philippines	0.890
Rwanda	0.749
South Africa	0.728
Sri Lanka	0.917
Suriname	-0.006
Thailand	0.656
Turkey	0.529
Yugoslavia	0.706

than 0.7. By and large, one can conclude that, despite the restriction that the slope coefficients are the same across countries, the relationship that has been postulated does not do badly in explaining the movements in the current account to exports ratio for the countries in this sample. It is possible that the fit could

have been improved if the basic economic parameters had been permitted to vary across countries, but even without such an extension the results turn out to be reasonably satisfactory, both on an average basis for the group of non-oil developing countries and for the individual countries making up the sample.

If the results are treated as relevant estimates for the “average” non-oil developing country in the sample, they suggest that this individual country was not completely powerless to adjust to exogenous shocks, since the authorities could have used an appropriate combination of demand-management and exchange rate policies to counter the effects of adverse changes in the other variables. For example, the estimates in Table 2 suggest that, *ceteris paribus*, it would require a depreciation of about $0.9x$ percent in the real effective exchange rate to keep the current account ratio unchanged in the face of a deterioration of x percent in the terms of trade.²⁰ This estimate assumes, of course, that the authorities are in a position to alter the *real* exchange rate by changing the nominal rate, an issue on which there is considerable dispute on both the theoretical and empirical levels.²¹ It could be that, owing to widespread indexation, this policy would not be feasible for some countries, since domestic factor prices would tend to “snap back” immediately following a devaluation, thereby leaving the real exchange rate unchanged. While the speed of this “pass-through” effect of a devaluation onto the domestic price level depends on the stance of economic policies as well as on the price responsiveness of domestic factor and product markets, the available evidence—admittedly for industrial countries—on the pass-through effect does not support the thesis of instantaneous reaction (Goldstein and Khan (1982 b)). Since there is no reason to believe that this conclusion does not carry over for the developing countries, particularly if appropriate supporting policies are implemented domestically, it seems reasonable to conclude that at least some portion of the current account effects of adverse international developments could be offset by a combination of a more flexible exchange rate policy and tighter demand-

²⁰By the same token, it would require a depreciation of 2.4 percent to offset a fall of 1 percentage point in the growth rate in industrial countries, and a depreciation of 0.6 percent to counter an increase of 1 percentage point in the foreign real interest rate.

²¹In this context, one must be careful to make a distinction between using nominal exchange rate adjustment to *restore* the equilibrium real exchange rate when it has moved out of line as against trying to change the equilibrium rate.

management policies designed to keep domestic inflation in check.

It should be stressed that, strictly speaking, the preceding estimates of the required depreciation in the real effective exchange rate that would be needed to offset a given change in foreign factors are relevant only for an individual country and cannot be applied to the group as a whole. An attempt by a large group of non-oil developing countries to depreciate their real exchange rates simultaneously would be likely to induce a deterioration in their terms of trade that would at least partly offset the positive effects of the exchange rate action. Since world demand for many primary commodities is not very responsive to price movements, this raises a policy dilemma for developing countries as a group: an exchange rate change will tend to be more effective for an individual developing country if its competitors refrain from similar action; obviously the beneficial effects of such a policy would be severely limited, or even eliminated entirely, if it were undertaken simultaneously by a large group of developing countries.

III. Conclusions

In view of the widespread interest in the balance of payments problems faced by non-oil developing countries during the 1970s, and their policy implications, the paucity of empirical work on this subject is quite surprising. This paper has attempted to further this discussion by examining the direct quantitative relationship between variations in the current account position and a set of factors that were assumed to be its main determinants, using pooled cross-section time-series analysis for a sample of 32 non-oil developing countries during the period 1973–80. In summary, the empirical tests here support the hypothesis that external factors (as represented by the secular decline in the terms of trade, the slowdown of economic growth in industrial countries, and the increase in foreign real interest rates) as well as domestic factors (captured by the fiscal deficit and appreciation in real effective exchange rates) were relevant in explaining the deterioration of current accounts of non-oil developing countries. Thus, the empirical results suggest the importance of exercising circumspection in attributing to any single cause the current account imbalances experienced by non-oil developing countries during the 1970s.

It has sometimes been asserted that the nature of a balance of payments stabilization program depends on the origin, or proximate cause, of disequilibrium. This view asserts that if a payments deficit is the result of excessively expansionary demand-management policies, the appropriate cure involves domestic demand restraint, whereas if the problem is caused by exogenous factors, such as a fall in the terms of trade, no adjustment is necessary and foreign financing should be provided. Since the results here indicate that both types of factor were at work during the 1970s, and as it is exceedingly difficult to separate the relative contributions of domestic and external factors to current account instability in a developing country (particularly in any *ex ante* sense), it would seem to make more practical sense to adopt an alternative view that has often been implicit in the work of the Fund. In this context, the question of whether a deficit ought principally to involve adjustment or financing should depend on whether the imbalance is viewed as permanent or temporary, irrespective of the origin of this imbalance. Such an approach has been stressed by, among others, Nowzad (1981), Guitián (1981), and Polak (1982). If developments that give rise to balance of payments difficulties are expected to be short lived and self-reversing, they may involve a need for temporary financing; permanent changes, on the other hand, necessitate adjustment of the basic supply/demand balance in the economy.

While one can argue that the slowdown of growth in industrial countries and the sharp rise in foreign real interest rates have been transitory phenomena and are likely to be reversed in the near future, the deterioration in the non-oil developing countries' terms of trade since 1974 appears to have been more in the nature of a long-term change. The terms of trade fell in five of the eight years 1974–81, and a further sharp decline of close to 12 percent is estimated to have taken place in 1982. Some financing of the deficits created by terms of trade changes did occur, but the situation also called for a substantial adjustment effort. In terms of the framework of this paper, evidence of insufficient adjustment in a number of developing countries is seen in the increase in fiscal deficits as well as in the way in which their real effective exchange rates appreciated during this period. For individual countries, suitable adjustment would have meant pursuing a more flexible exchange rate policy, supplemented by the application of a broad range of demand-management policies. Some countries, notably those among the group that are classified as major ex-

porters of manufactures or "newly industrialized countries," did adopt such a strategy with considerable success. However, the beginning of the 1980s also found a large number of non-oil developing countries experiencing increased current account deficits resulting not only from adverse international developments but also from domestic developments during the decade.

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